

Complete Publication List

Oliver Gutsche
September 01, 2026

1. Evidence for the dead-cone effect in bottom quark initiated jets produced in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.29412](#) | 2026-08-29
2. Search for scalar leptoquarks produced via muon-quark scattering in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.28377](#) | 2026-08-28
3. Search for top quark flavor-changing neutral currents in multilepton final states using an effective field theory approach in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.19547](#) | 2026-08-20
4. Observation of an excess at the top quark pair production threshold in the single-lepton channel
Gevorgyan, Arzunik et al.
[arXiv:2608.19546](#) | 2026-08-20
5. Performance of heavy-flavour jet identification in the CMS high-level trigger in proton-proton collisions at $\sqrt{s} = 13.6$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.18331](#) | 2026-08-18
6. Search for anomalous couplings in WW and WZ production with single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.18245](#) | 2026-08-18
7. Search for a resonance in events with four top quarks decaying into two leptons and jets in proton-proton collisions
Gevorgyan, Arzunik et al.
[arXiv:2608.10148](#) | 2026-08-10
8. Search for resonant production of lepton-enriched semivisible jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.05323](#) | 2026-08-05
9. Combination of vector boson scattering measurements in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
Gevorgyan, Arzunik et al.
[arXiv:2608.01289](#) | 2026-08-02
10. Search for a top-philic Z' boson decaying into a $t\bar{t}$ pair in a final state with jets and an electron or muon in proton-proton collisions at $\sqrt{s} = 13.6$ TeV
Hayrapetyan, Aram et al.
[arXiv:2608.01251](#) | 2026-08-02

11. A measurement of the Higgs boson mass in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.28396](#) | 2026-07-30

12. Measurement of the fragmentation properties of jets containing $\Upsilon(nS)$ mesons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.25653](#) | 2026-07-28

13. Off-shell Higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width

Gevorgyan, Arzunik et al.

[arXiv:2607.23352](#) | 2026-07-25

14. Measurement of the $\gamma\gamma \rightarrow \tau\tau$ cross section and constraints on the anomalous magnetic moment of the τ lepton in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2607.20148](#) | 2026-07-22

15. Study of χ_c production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Belyaev, Andrey et al.

[arXiv:2607.17847](#) | 2026-07-20

16. Evidence for sequential $\Upsilon(nS)$ suppression in light ion collisions

Belyaev, Andrey et al.

[arXiv:2607.13758](#) | 2026-07-15

17. Exploring the origin of D^0 meson elliptic flow in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using event shape engineering

Gevorgyan, Arzunik et al.

[arXiv:2607.11017](#) | 2026-07-13

18. Search for dark matter in a signature with a four-prong large-radius jet in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.11016](#) | 2026-07-13

19. Search for soft unclustered energy patterns in proton-proton collisions at $\sqrt{s} = 13$ TeV using data scouting

Gevorgyan, Arzunik et al.

[arXiv:2607.09621](#) | 2026-07-10

20. Search for nonresonant triple Higgs boson production in the final state with six bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.05145](#) | 2026-07-06

21. Study of ZZ and ZH production in the $b\bar{b}\tau\tau$ final state and search for high-mass spin-0 and spin-1 resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.01589](#) | 2026-07-02

22. Search for physics beyond the standard model in four and three top quark production events using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2606.30862](#) | 2026-06-29

23. Precision luminosity measurement in proton-proton collisions at a center-of-mass energy of 13 TeV with the CMS detector at the Large Hadron Collider

Gevorgyan, Arzunik et al.

[arXiv:2606.26832](#) | 2026-06-25

24. Search for long-lived particles decaying into muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV using data scouting

Gevorgyan, Arzunik et al.

[arXiv:2606.25840](#) | 2026-06-24

25. Observation of electroweak production of pairs of Z bosons in proton-proton collisions at 13 TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.20558](#) | 2026-06-18

26. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and tau leptons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.03510](#) | 2026-06-02

27. Search for single production of a vector-like B' quark decaying to a top quark and a W boson in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tauqeer, Komal et al.

[arXiv:2606.01423](#) | 2026-05-31

28. Dependence of two-particle azimuthal correlations on the forward rapidity gap width in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Hayrapetyan, Aram et al.

[arXiv:2606.00171](#) | 2026-05-29

29. Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.27875](#) | 2026-05-27

30. Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.26462](#) | 2026-05-26

31. Search for low-mass resonances decaying to $\tau\tau$ and measurement of the $Y \rightarrow \tau\tau$ decay in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.25103](#) | 2026-05-24

32. Studies of $Z \rightarrow 4\ell$ decays in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Tumasyan, Armen et al.

[arXiv:2605.25158](#) | 2026-05-24

33. Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.19144](#) | 2026-05-18

34. Search for the nonresonant production of a pair of additional Higgs bosons in the Type-X two-Higgs-doublet model in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.16655](#) | 2026-05-15

35. Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach

Hayrapetyan, Aram et al.

[arXiv:2605.15023](#) | 2026-05-14

36. Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.14245](#) | 2026-05-14

37. First measurements of vector boson scattering in $W^\pm W^\pm$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.15396](#) | 2026-05-14

38. Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.13614](#) | 2026-05-13

39. Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.13649](#) | 2026-05-13

40. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

41. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

42. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

43. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

44. Measurement of the top quark pair production cross section in PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV

Belyaev, Andrey et al.

[JHEP 08 \(2026\) 054](#) | [arXiv:2604.27091](#) | [DOI:10.1007/JHEP08\(2026\)054](#) | 2026-04-29

45. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

46. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.
[arXiv:2604.25538](#) | 2026-04-28

47. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.
[arXiv:2604.25678](#) | 2026-04-28

48. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.24531](#) | 2026-04-27

49. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.20999](#) | 2026-04-22

50. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.17564](#) | 2026-04-19

51. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.13320](#) | 2026-04-14

52. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.14236](#) | 2026-04-14

53. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
JHEP 08 (2026) 098 | [arXiv:2604.10732](#) | [DOI:10.1007/JHEP08\(2026\)098](#) | 2026-04-12

54. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
Phys.Lett.B 880 (2026) 140812 | [arXiv:2604.10729](#) | [DOI:10.1016/j.physletb.2026.140812](#) | 2026-04-12

55. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.
[arXiv:2604.09809](#) | 2026-04-10

56. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.
[arXiv:2604.05814](#) | 2026-04-07

57. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
Phys.Lett.B 879 (2026) 140671 | [arXiv:2604.05996](#) | [DOI:10.1016/j.physletb.2026.140671](#) | 2026-04-07

58. Evidence of $ZZ\gamma$ Production and Observation of $4\ell\gamma$ in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 137 \(2026\) 091805](#) | [arXiv:2604.02594](#) | [DOI:10.1103/gdbv-wnpp](#) | 2026-04-03

59. Search for Higgs boson pair production in the $bbWW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s}=13.6\text{ TeV}$

Gevorgyan, Arzunik et al.

[JHEP 08 \(2026\) 019](#) | [arXiv:2604.02127](#) | [DOI:10.1007/JHEP08\(2026\)019](#) | 2026-04-02

60. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

61. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

62. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

63. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 248](#) | [arXiv:2603.22233](#) | [DOI:10.1007/JHEP07\(2026\)248](#) | 2026-03-23

64. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 054](#) | [arXiv:2603.19963](#) | [DOI:10.1007/JHEP07\(2026\)054](#) | 2026-03-20

65. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Belyaev, Andrey et al.

[arXiv:2603.14385](#) | 2026-03-15

66. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s}=5.02\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 08 \(2026\) 223](#) | [arXiv:2603.13592](#) | [DOI:10.1007/JHEP08\(2026\)223](#) | 2026-03-13

67. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2603.11035](#) | 2026-03-11

68. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2603.03689](#) | 2026-03-04

69. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 114 \(2026\) 032005](#) | [arXiv:2603.03250](#) | [DOI:10.1103/q2mh-7mhj](#) | 2026-03-03

70. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.

[arXiv:2602.23991](#) | 2026-02-27

71. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 112005](#) | [arXiv:2602.20477](#) | [DOI:10.1103/bwqs-vgqn](#) | 2026-02-24

72. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.

[Phys.Lett.B 880 \(2026\) 140679](#) | [arXiv:2602.21325](#) | [DOI:10.1016/j.physletb.2026.140679](#) | 2026-02-24

73. Observation of the Jet Diffusion Wake Using Dijets in Heavy-Ion Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 137 \(2026\) 071902](#) | [arXiv:2602.19431](#) | [DOI:10.1103/g49y-8cjl](#) | 2026-02-23

74. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

75. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

76. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 89 \(2026\) 087801](#) | [arXiv:2602.18611](#) | [DOI:10.1088/1361-6633/ae72c4](#) | 2026-02-20

77. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

78. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

79. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks

Giannakou, Anna et al.

[\(2026\) 349](#) | [DOI:10.1109/ICNC68183.2026.11416957](#) | 2026

80. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

81. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

82. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[Phys.Rev.D 114 \(2026\) 012005](#) | [arXiv:2602.10270](#) | [DOI:10.1103/36cv-vst3](#) | 2026-02-10

83. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

84. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

85. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

86. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2026\) 050](#) | [arXiv:2602.00273](#) | [DOI:10.1007/JHEP08\(2026\)050](#) | 2026-01-30

87. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2026\) 187](#) | [arXiv:2601.21574](#) | [DOI:10.1007/JHEP06\(2026\)187](#) | 2026-01-29

88. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 878 \(2026\) 140498](#) | [arXiv:2601.20063](#) | [DOI:10.1016/j.physletb.2026.140498](#) | 2026-01-27

89. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s}=13.6$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 106](#) | [arXiv:2601.20023](#) | [DOI:10.1007/JHEP07\(2026\)106](#) | 2026-01-27

90. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 86 \(2026\) 901](#) | [arXiv:2601.20011](#) | [DOI:10.1140/epjc/s10052-026-15699-9](#) | 2026-01-27

91. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

92. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 102](#) | [arXiv:2601.17576](#) | [DOI:10.1007/JHEP07\(2026\)102](#) | 2026-01-24

93. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

94. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

95. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2026\) 049](#) | [arXiv:2601.14102](#) | [DOI:10.1007/JHEP08\(2026\)049](#) | 2026-01-20

96. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 005](#) | [arXiv:2601.12583](#) | [DOI:10.1007/JHEP07\(2026\)005](#) | 2026-01-18

97. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092016](#) | [arXiv:2601.11330](#) | [DOI:10.1103/2m87-3dlr](#) | 2026-01-16

98. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092014](#) | [arXiv:2601.05168](#) | [DOI:10.1103/rny9-zby1](#) | 2026-01-08

99. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 879 \(2026\) 140534](#) | [arXiv:2601.05362](#) | [DOI:10.1016/j.physletb.2026.140534](#) | 2026-01-08

100. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 119](#) | [arXiv:2601.00183](#) | [DOI:10.1007/JHEP07\(2026\)119](#) | 2026-01-01

101. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

102. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[Mach.Learn.Sci.Tech. 7 \(2026\) 045008](#) | [arXiv:2512.20395](#) | [DOI:10.1088/2632-2153/ae7d87](#) | 2025-12-23

103. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13$ TeV using the beam and helicity bases

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

104. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

105. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2026\) 020](#) | [arXiv:2512.00502](#) | [DOI:10.1007/JHEP05\(2026\)020](#) | 2025-11-29

106. Search for Light Pseudoscalar Bosons, Pair-Produced in Higgs Boson Decays in the Four-Electron Final State in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 136 \(2026\) 181807](#) | [arXiv:2511.19563](#) | [DOI:10.1103/r53x-8fqp](#) | 2025-11-24

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[Phys.Rev.D 113 \(2026\) 092009](#) | [arXiv:2511.16394](#) | [DOI:10.1103/w3b5-92ls](#) | 2025-11-20

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Chekhovsky, Vladimir et al.

[JHEP 07 \(2026\) 008](#) | [arXiv:2511.11853](#) | [DOI:10.1007/JHEP07\(2026\)008](#) | 2025-11-14

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Chekhovsky, Vladimir et al.

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115. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

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117. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

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Hayrapetyan, Aram et al.

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[arXiv:2510.07527](#) | 2025-10-08

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127. Vector boson scattering and anomalous quartic couplings in final states with $\ell\nu qq$ or $\ell\ell qq$ plus jets using proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

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[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

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172. Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

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181. Evidence for Similar Collectivity of High Transverse-Momentum Particles in p-Pb and Pb-Pb Collisions
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186. Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
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190. Determination of the strong coupling and its running from measurements of inclusive jet production

Chekhovsky, Vladimir et al.

[Phys.Lett.B 868 \(2025\) 139651](#) | [arXiv:2412.16665](#) | [DOI:10.1016/j.physletb.2025.139651](#) | 2024-12-21

191. Measurement of the W boson decay branching fraction ratio $\mathcal{B}(W \rightarrow c\bar{q})/\mathcal{B}(W \rightarrow q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 868 \(2025\) 139754](#) | [arXiv:2412.16296](#) | [DOI:10.1016/j.physletb.2025.139754](#) | 2024-12-20

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Hayrapetyan, Aram et al.

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Chekhovsky, Vladimir et al.

[Nature 652 \(2026\) 321](#) | [arXiv:2412.13872](#) | [DOI:10.1038/s41586-026-10168-5](#) | 2024-12-18

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Hayrapetyan, Aram et al.

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199. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at

$\sqrt{s} = 13 \text{ TeV}$

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200. Search for rare decays of the Z and Higgs bosons to a J/ψ or $\psi(2S)$ meson and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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Hayrapetyan, Aram et al.

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203. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2025\) 006](#) | [arXiv:2410.21137](#) | [DOI:10.1007/JHEP06\(2025\)006](#) | 2024-10-28

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242. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

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250. Search for new physics in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV
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252. Girth and groomed radius of jets recoiling against isolated photons in lead-lead and proton-proton collisions at $\sqrt{s_{NN}}=5.02$ TeV
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253. Search for New Resonances Decaying to Pairs of Merged Diphotons in Proton-Proton Collisions at $\sqrt{s}=13$ TeV
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255. Performance of CMS muon reconstruction from proton-proton to heavy ion collisions
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[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26
256. Measurement of multijet azimuthal correlations and determination of the strong coupling in proton-proton collisions at $\sqrt{s}=13$ TeV
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257. Search for Higgs boson pair production with one associated vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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260. Searches for Pair-Produced Multijet Resonances Using Data Scouting in Proton-Proton Collisions at $\sqrt{s}$ =13TeV

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[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03

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262. Measurement of the production cross section of a Higgs boson with large transverse momentum in its decays to a pair of τ leptons in proton-proton collisions at $\sqrt{s}$ =13 TeV

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[Phys.Lett.B 857 \(2024\) 138964](#) | [arXiv:2403.20201](#) | [DOI:10.1016/j.physletb.2024.138964](#) | 2024-03-29

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264. Enriching the physics program of the CMS experiment via data scouting and data parking

Hayrapetyan, Aram et al.

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267. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s}$ = 13 TeV

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Hayrapetyan, Aram et al.

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Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

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Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

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[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

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[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

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Hayrapetyan, Aram et al.

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282. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

283. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

284. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

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[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

286. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

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[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

288. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

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[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

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[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

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[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

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[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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295. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^{+} \mu^{-}$ and $B^{\pm} \rightarrow K^{\pm} e^{+} e^{-}$ decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

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[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

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[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

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[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

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[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

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Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

303. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

304. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

305. Search for charged-lepton flavor violation in the production and decay of top quarks using trilepton final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

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Gutsche, Oliver et al.

[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

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[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

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Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

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[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

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[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

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[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

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[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

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[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-10-18

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[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

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[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

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[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

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[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

321. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

323. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

324. Inclusive and differential cross section measurements of $t\bar{t}b\bar{b}$ production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

325. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

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[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

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[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

328. Luminosity determination using Z boson production at the CMS experiment

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[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

329. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

330. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

331. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

332. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

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333. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

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[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

334. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

335. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

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[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

336. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

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[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

337. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

338. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

339. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

340. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

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341. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

342. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

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[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

343. New Structures in the $J/\psi/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

344. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

345. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

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[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

346. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

347. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

348. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

349. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

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Balcas, Justas et al.

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Tumasyan, Armen et al.

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Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

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[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

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[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

359. Search for a vector-like quark $T \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

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[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

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362. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

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[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

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367. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

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Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

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[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

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[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

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Tumasyan, Armen et al.

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[arXiv:2203.09945](#) | 2022-03-15

435. Search for new particles in an extended Higgs sector with four b quarks in the final state at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

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[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

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Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

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[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

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[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

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[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

448. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

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[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

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[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

452. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

453. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

454. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s}=13$ TeV collected with a dedicated high-rate data stream

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[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

455. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

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[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

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[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

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[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

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[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

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461. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

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[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

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[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

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[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

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[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

466. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

467. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

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[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

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[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

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[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

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[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

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[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

473. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

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[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

475. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

476. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

477. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

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[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

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[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

480. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

481. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

482. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

483. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

484. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

485. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

486. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

487. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

488. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

489. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

490. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

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[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

491. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

492. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

493. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

494. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

495. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

496. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

497. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

498. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

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Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

500. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

501. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

502. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

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[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

503. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

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[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

504. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

505. Measurements of the $pp \rightarrow W^{\pm}\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

506. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

507. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

508. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

509. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

510. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

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[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

511. Search for W' bosons decaying to a top and a bottom quark at $s=13\text{TeV}$ in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

512. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

513. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

514. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{\text{NN}}}=5.02$ TeV

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Sirunyan, Albert M et al.

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684. Search for new physics in top quark production in dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 79 \(2019\) 886](#) | [arXiv:1903.11144](#) | [DOI:10.1140/epjc/s10052-019-7387-y](#) | 2019-03-26

685. Search for a low-mass $\tau^+\tau^-$ resonance in association with a bottom quark in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2019\) 210](#) | [arXiv:1903.10228](#) | [DOI:10.1007/JHEP05\(2019\)210](#) | 2019-03-25

686. Search for supersymmetry in final states with photons and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112002](#) | [arXiv:1903.06973](#) | [DOI:10.1103/PhysRevD.100.112002](#) | 2019-03-16

688. Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s}=13$ TeV using the CMS detector

Sirunyan, Albert M et al.

[JINST 14 \(2019\) P07004](#) | [arXiv:1903.06078](#) | [DOI:10.1088/1748-0221/14/07/P07004](#) | 2019-03-14

689. Search for charged Higgs bosons in the $H^\pm \rightarrow \tau^\pm \nu_\tau$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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[Eur.Phys.J.C 80 \(2020\) 43](#) | [arXiv:1903.04040](#) | [DOI:10.1140/epjc/s10052-019-7585-7](#) | 2019-03-10

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Sirunyan, Albert M et al.

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Aaboud, Morad et al.

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[Phys.Lett.B 799 \(2019\) 135049](#) | [arXiv:1902.03603](#) | [DOI:10.1016/j.physletb.2019.135049](#) | 2019-02-10

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767. Search for an L_μ - L_τ gauge boson using $Z \rightarrow 4\mu$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 345](#) | [arXiv:1808.03684](#) | [DOI:10.1016/j.physletb.2019.01.072](#) | 2018-08-10

768. Search for pair-produced resonances decaying to quark pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 112014](#) | [arXiv:1808.03124](#) | [DOI:10.1103/PhysRevD.98.112014](#) | 2018-08-09

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Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 092011](#) | [arXiv:1808.03078](#) | [DOI:10.1103/PhysRevD.98.092011](#) | 2018-08-09

770. Evidence for the associated production of a single top quark and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 121 \(2018\) 221802](#) | [arXiv:1808.02913](#) | [DOI:10.1103/PhysRevLett.121.221802](#) | 2018-08-08

771. Search for resonances in the mass spectrum of muon pairs produced in association with b quark jets in proton-proton collisions at $\sqrt{s}=8$ and 13 TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2018\) 161](#) | [arXiv:1808.01890](#) | [DOI:10.1007/JHEP11\(2018\)161](#) | 2018-08-06

772. Search for production of Higgs boson pairs in the four b quark final state using large-area jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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773. Search for narrow $H\gamma$ resonances in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03

774. Search for heavy resonances decaying into two Higgs bosons or into a Higgs boson and a W or Z boson in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 051](#) | [arXiv:1808.01365](#) | [DOI:10.1007/JHEP01\(2019\)051](#) | 2018-08-03

775. Search for a W' boson decaying to a τ lepton and a neutrino in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 792 \(2019\) 107](#) | [arXiv:1807.11421](#) | [DOI:10.1016/j.physletb.2019.01.069](#) | 2018-07-30

776. Searches for pair production of charginos and top squarks in final states with two oppositely charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

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777. Search for the Higgs boson decaying to two muons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 122 \(2019\) 021801](#) | [arXiv:1807.06325](#) | [DOI:10.1103/PhysRevLett.122.021801](#) | 2018-07-17

778. Search for dark matter particles produced in association with a top quark pair at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 011803](#) | [arXiv:1807.06522](#) | [DOI:10.1103/PhysRevLett.122.011803](#) | 2018-07-17

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Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 113](#) | [arXiv:1807.05974](#) | [DOI:10.1007/JHEP11\(2018\)113](#) | 2018-07-16

780. Measurement of inclusive and differential Higgs boson production cross sections in the diphoton decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

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Sirunyan, Albert M et al.

[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

782. Study of the underlying event in top quark pair production in pp collisions at 13 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

783. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 172](#) | [arXiv:1807.02826](#) | [DOI:10.1007/JHEP11\(2018\)172](#) | 2018-07-08

784. Search for supersymmetry in events with a τ lepton pair and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 151](#) | [arXiv:1807.02048](#) | [DOI:10.1007/JHEP11\(2018\)151](#) | 2018-07-05

785. Measurement of differential cross sections for inclusive isolated-photon and photon+jets production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02

786. Measurement of charged particle spectra in minimum-bias events from proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 697](#) | [arXiv:1806.11245](#) | [DOI:10.1140/epjc/s10052-018-6144-y](#) | 2018-06-29

787. Search for heavy Majorana neutrinos in same-sign dilepton channels in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 122](#) | [arXiv:1806.10905](#) | [DOI:10.1007/JHEP01\(2019\)122](#) | 2018-06-28

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Sirunyan, Albert M. et al.

[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28

789. Search for the decay of a Higgs boson in the $\ell\ell\gamma$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 152](#) | [arXiv:1806.05996](#) | [DOI:10.1007/JHEP11\(2018\)152](#) | 2018-06-15

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[Phys.Lett.B 791 \(2019\) 96](#) | [arXiv:1806.05246](#) | [DOI:10.1016/j.physletb.2018.12.073](#) | 2018-06-13

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Sirunyan, Albert M. et al.

[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13

792. Search for dark matter produced in association with a Higgs boson decaying to $\gamma\gamma$ or $\tau^+\tau^-$ at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 09 \(2018\) 046](#) | [arXiv:1806.04771](#) | [DOI:10.1007/JHEP09\(2018\)046](#) | 2018-06-12

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[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11

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Sirunyan, Albert M. et al.

[JHEP 07 \(2018\) 115](#) | [arXiv:1806.03472](#) | [DOI:10.1007/JHEP07\(2018\)115](#) | 2018-06-09

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[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

797. Search for narrow and broad dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on dark matter mediators and other new particles

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 130](#) | [arXiv:1806.00843](#) | [DOI:10.1007/JHEP08\(2018\)130](#) | 2018-06-03

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 701](#) | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

799. Angular analysis of the decay $B^+ \rightarrow K^+ \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Rev.D 98 \(2018\) 112011](#) | [arXiv:1806.00636](#) | [DOI:10.1103/PhysRevD.98.112011](#) | 2018-06-02

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[Phys.Lett.B 788 \(2019\) 7](#) | [arXiv:1806.00408](#) | [DOI:10.1016/j.physletb.2018.10.056](#) | 2018-06-01

801. Search for beyond the standard model Higgs bosons decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV

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[JHEP 08 \(2018\) 113](#) | [arXiv:1805.12191](#) | [DOI:10.1007/JHEP08\(2018\)113](#) | 2018-05-30

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[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

803. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

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[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

805. Measurement of nuclear modification factors of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

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807. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

808. Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 042](#) | [arXiv:1805.06013](#) | [DOI:10.1007/JHEP11\(2018\)042](#) | 2018-05-15

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Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 161](#) | [arXiv:1805.05145](#) | [DOI:10.1007/JHEP10\(2018\)161](#) | 2018-05-14

810. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state of two muons and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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[Phys.Rev.Lett. 121 \(2018\) 062002](#) | [arXiv:1805.04736](#) | [DOI:10.1103/PhysRevLett.121.062002](#) | 2018-05-12

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[JHEP 08 \(2018\) 177](#) | [arXiv:1805.04758](#) | [DOI:10.1007/JHEP08\(2018\)177](#) | 2018-05-12

813. Measurement of prompt $\psi(2S)$ production cross sections in proton-lead and proton-proton collisions at $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

814. Measurement of the top quark mass with lepton+jets final states using pp collisions at $\sqrt{s}=13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 891](#) | [arXiv:1805.01428](#) | [DOI:10.1140/epjc/s10052-018-6332-9](#) | 2018-05-03

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Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

816. Search for disappearing tracks as a signature of new long-lived particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

817. Measurement of differential cross sections for Z boson production in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

818. Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s}= 13$ TeV

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[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

819. Search for ttH production in the $H \rightarrow b\bar{b}$ decay channel with leptonic tt decays in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 03 \(2019\) 026](#) | [arXiv:1804.03682](#) | [DOI:10.1007/JHEP03\(2019\)026](#) | 2018-04-10

820. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation

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[arXiv:1804.03983](#) | 2018-04-09

821. Measurements of Higgs boson properties in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 11 \(2018\) 185](#) | [arXiv:1804.02716](#) | [DOI:10.1007/JHEP11\(2018\)185](#) | 2018-04-08

822. Observation of ttH production

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[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

823. Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 127](#) | [arXiv:1804.01939](#) | [DOI:10.1007/JHEP06\(2018\)127](#) | 2018-04-05

824. Search for a heavy right-handed W boson and a heavy neutrino in events with two same-flavor leptons and two jets at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

825. Search for high-mass resonances in final states with a lepton and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 128](#) | [arXiv:1803.11133](#) | [DOI:10.1007/JHEP06\(2018\)128](#) | 2018-03-29

826. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

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[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

827. Measurement of differential cross sections for the production of top quark pairs and of additional jets in lepton+jets events from pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

828. Search for new physics in dijet angular distributions using proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on dark matter and other models

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[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

829. Search for ttH production in the all-jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

830. Search for additional neutral MSSM Higgs bosons in the $\tau\tau$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

831. Search for high-mass resonances in dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

832. Evidence for associated production of a Higgs boson with a top quark pair in final states with electrons, muons, and hadronically decaying τ leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

833. Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

834. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

835. Search for a heavy resonance decaying into a Z boson and a vector boson in the $\nu\nu q\bar{q}$ final state

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

836. Search for third-generation scalar leptoquarks decaying to a top quark and a τ lepton at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

837. Jet properties in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

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[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

838. Search for a heavy resonance decaying to a pair of vector bosons in the lepton plus merged jet final state at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

839. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

840. Measurement of the Λ_b polarization and angular parameters in $\Lambda_b \rightarrow J/\psi \Lambda$ decays from pp collisions at $\sqrt{s} = 7$ and 8 TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

841. Search for heavy neutral leptons in events with three charged leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

842. Measurement of the inelastic proton-proton cross section at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

843. Search for natural and split supersymmetry in proton-proton collisions at $\sqrt{s}=13$ TeV in final states with jets and missing transverse momentum

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

844. Search for single production of vector-like quarks decaying to a b quark and a Higgs boson

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

845. Search for lepton-flavor violating decays of heavy resonances and quantum black holes to $e\mu$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

846. Comparing transverse momentum balance of b jet pairs in pp and PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

847. Search for dark matter in events with energetic, hadronically decaying top quarks and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 027](#) | [arXiv:1801.08427](#) | [DOI:10.1007/JHEP06\(2018\)027](#) | 2018-01-25

848. Observation of Medium-Induced Modifications of Jet Fragmentation in Pb-Pb Collisions at $\sqrt{s_{NN}}= 5.02$ TeV Using Isolated Photon-Tagged Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

849. Combined search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 160](#) | [arXiv:1801.03957](#) | [DOI:10.1007/JHEP03\(2018\)160](#) | 2018-01-11

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1085. Search for heavy resonances decaying to two Higgs bosons in final states containing four b quarks

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Lett.B 759 \(2016\) 9](#) | [arXiv:1512.08002](#) | [DOI:10.1016/j.physletb.2016.05.033](#) | 2015-12-25

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Garzoglio, G. et al.

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Apyan, A. et al.

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Balcas, J. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 062030](#) | [DOI:10.1088/1742-6596/664/6/062030](#) | 2015-12-23

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Balcas, J. et al.

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1167. Measurement of the Z boson differential cross section in transverse momentum and rapidity in proton-proton collisions at 8 TeV

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[Phys.Lett.B 749 \(2015\) 337](#) | [arXiv:1502.07400](#) | [DOI:10.1016/j.physletb.2015.07.053](#) | 2015-02-25

1182. Search for Physics Beyond the Standard Model in Events with Two Leptons, Jets, and Missing Transverse Momentum in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 124](#) | [arXiv:1502.06031](#) | [DOI:10.1007/JHEP04\(2015\)124](#) | 2015-02-20

1183. Measurement of the $Z\gamma$ Production Cross Section in pp Collisions at 8 TeV and Search for Anomalous Triple Gauge Boson Couplings

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 164](#) | [arXiv:1502.05664](#) | [DOI:10.1007/JHEP04\(2015\)164](#) | 2015-02-19

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 115 \(2015\) 012301](#) | [arXiv:1502.05382](#) | [DOI:10.1103/PhysRevLett.115.012301](#) | 2015-02-18

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 237](#) | [arXiv:1502.05387](#) | [DOI:10.1140/epjc/s10052-015-3435-4](#) | 2015-02-18

1186. Search for Narrow High-Mass Resonances in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV Decaying to a Z and a Higgs Boson

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 255](#) | [arXiv:1502.04994](#) | [DOI:10.1016/j.physletb.2015.07.011](#) | 2015-02-17

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 302](#) | [arXiv:1502.04785](#) | [DOI:10.1140/epjc/s10052-015-3491-9](#) | 2015-02-16

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 191802](#) | [arXiv:1502.04155](#) | [DOI:10.1103/PhysRevLett.114.191802](#) | 2015-02-13

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 251](#) | [arXiv:1502.02485](#) | [DOI:10.1140/epjc/s10052-015-3454-1](#) | 2015-02-09

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Khachatryan, Vardan et al.

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[JINST 10 \(2015\) P06005](#) | [arXiv:1502.02701](#) | [DOI:10.1088/1748-0221/10/06/P06005](#) | 2015-02-09

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Lett.B 748 \(2015\) 144](#) | [arXiv:1501.05566](#) | [DOI:10.1016/j.physletb.2015.06.070](#) | 2015-01-22

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[Phys.Lett.B 746 \(2015\) 79](#) | [arXiv:1411.2646](#) | [DOI:10.1016/j.physletb.2015.04.042](#) | 2014-11-10

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Khachatryan, Vardan et al.

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[Phys.Lett.B 740 \(2015\) 250](#) | [arXiv:1406.0113](#) | [DOI:10.1016/j.physletb.2016.04.010](#) | 2014-05-31

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Khachatryan, Vardan et al.

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1268. Search for New Physics in the Multijet and Missing Transverse Momentum Final State in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

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